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Title: Some silvicultural impacts of Phellinus weirii on a managed forest in the interior Coast Range of Oregon

Abstract approved: Alan B. Berg

Phellinus weirii, the cause of laminated root rot, is considered the most destructive root rot of Douglas-fir, but little is known about the damage it causes to a forest. The objective of this study was to determine silvicultural impacts of P. weirii on a managed forest.

A survey located and mapped all P. weirii infection centers on 195 hectares of the George T. Gerlinger Experimental Forest in the interior Coast Range of Oregon. Trees within infection centers were examined for P. weirii, their diameters measured, and their locations mapped.

The survey located 27 infection centers, ranging from 0.02 hectare with 5 trees to 0.71 hectares with 148 trees. The total area infected was 7.13 hectares. Within these centers were 1732 Douglas-fir infected by P. weirii, with a total volume of 2924.9 cubic meters (513,780 board feet). In one year (1978), volume loss of Douglas-fir due to P. weirii was 457.75 cubic meters. Salvage records show variation in damage on the whole forest from 1957 to 1978. The fungus was cultured from dead trees in five of the infection centers and showed isolate differences between and within centers

Also included in the survey were two areas, the Upper and Lower *Phellinus* plots, which were established in 1961 to study this disease. Records were kept of tree deaths from 1961 to 1972. These areas were reinventoried for recent kills in 1978. The location of *P. weirii* killed trees on the Upper and Lower plots from 1961 to 1978 showed no evidence for uniform, annual expansion of infection centers.

The fungus varied in its damage over 17 years on the Upper and Lower *Phellinus* plots. For instance, on one subplot it caused 54 percent of the mortality from 1962 to 1967, no mortality from 1968 to 1972, and all the deaths from 1973 to 1978.

Management proposals by the U.S. Forest Service and the Canadian Foresty Service for infected stands were inadequate for the forest studied here. Guidelines are proposed for management of infected stands.

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Some Silvicultural Impacts of *Phellinus weirii* on a Managed Forest in the Interior Coast Range of Oregon

by

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Some silvicultural impacts of *Phellinus weirii* on a managed forest in the interior Coast Range of Oregon.

Introduction

To the people of the eighteenth and nineteenth centuries, North America was almost magical, described by some as a garden of Eden, by others as a land of hostile Indians. As explorers became acquainted with the region now known as the Pacific Northwest, one dominant feature was its vast forests. These forests attracted both trappers for the resident animals and botanists for the trees.

One of the more attractive trees in the Pacific Northwest was Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco), first discovered in 1792 by Archibald Menzies, who visited the Northwest with Captain Vancouver. Lewis and Clark also mention in their journals large fir trees which probably were Douglas-fir (Thwaites 1905), but it was not until the travels of the Scottish botanist David Douglas from 1824 to 1827 that samples of this species were collected and a taxonomic description filed. In commenting on the large trees of those virgin forests, Douglas noted a windthrown Douglas-fir to be 42 meters tall, with a 46 centimeter top diameter, and a circumference of 6 meters (Royal Horticulture Society 1914).

Forests of such trees supplied the gold mines of California with mine timbers in the 1840s (Vaughan 1959), and by 1910 provided raw material for over 1500 sawmills in Oregon and Washington (Cornwall 1910). At the turn of the century, some considered Douglas-fir one of the more important species in the land:

> No other important commercial tree in America is so widely distributed as Douglas fir, and it is safe to say that none which has been introduced into Europe has attracted so much attention among foresters. The immense size which it attains in portions of its range, its high commercial value, rapid growth, and above all the ease with which it lends to silvicultural management all combine to place it among the most useful of the world's forest trees. There is little danger that the tree will ever cease to be an important part of the forests within its range.
>
> -- Frothingham (1908)

Douglas-fir has retained its importance. The coastal region of Oregon and Washington comprises 4.6 million hectares and produces a net volume of growing stock of 4,448 million cubic meters (U.S. Department of Agriculture 1978). The forest products industry employs 80,000 workers in Oregon alone (Brodie et al. 1978).

When white settlers came to the Pacific Northwest, the virgin forests seemed so expansive that no one thought they would ever be depleted. This belief and market conditions led to destructive logging. Timber was considered a nonrenewable resource (Munger 1911). This attitude caused some to fear a desolation of the Nation's forest lands: Rohan (1929) stated that 50 percent of the western forests had been logged and little of the land replanted. But in fact regeneration and management of Pacific Northwest forests had begun in the early 1900s.

When the foresters looked at mature stands of Douglas-fir, they found that young Douglas-fir were absent from the understory, but hemlock, grand fir, and western red cedar were present. Such an understory would result in the replacement of the stands of Douglas-fir by an "irregular forest of inferior species (Munger 1912)." This change in

dominant species was unfavorably regarded because "the universal aim of silviculture is to get the most and best timber per unit area in the shortest possible time, '[and] on the humid mountain slopes of Washington and Oregon no tree is better suited to achieve this ideal than Douglas-fir (Frothingham 1908)." So, as old-growth stands were cut, foresters chose to establish Douglas-fir in the second-growth stands.

With this in mind, the management objective of replacing old-growth forests with Douglas-fir became rather simple. For pure stands of Douglas-fir it was recommended to cut clean (clearcut) and burn to expose mineral soil, the preferred seed bed for Douglas-fir. For mixed stands, it was recommended that all species be cut except enough Douglas-fir to serve as seed trees (Frothingham 1908). Douglas-fir became the more favored species and the dominant species in the forest.

This change in species composition may have awakened an organism native to the Douglas-fir ecosystem-- the fungus Phellinus weirii (Murr.) Gilbertson. Except for P. pini, pathogens of western forest trees received little attention until the late 1940s. Perhaps because the forests were so vast forest managers thought in the broadest of terms or took a familiar stand and applied its condition to the whole forest, thus overlooking the impact of fungi (Meinecke 1916). However, by 1940 windthrow of Douglas-fir appeared to be increasing, and small centers of mortality filling with brush were noticed in the forest. At first this was attributed to the Douglas-fir bark beetle, but by 1954 P. weirii, the cause of laminated root rot, was given credit for the damage.

Phellinus weirii is considered "the most destructive root rot of Douglas-fir (Childs and Nelson 1971)," and has been estimated to cause an annual loss of almost 1 million cubic meters in Douglas-fir west of the Cascade Mountains (Childs and Shea 1967). With an industry and region so dependent on Douglas-fir, P. weirii assumes importance beyond that of most forest pathogens.

The objective of this study was to determine the silvicultural impacts of Phellinus weirii on a managed forest.

Literature Review

Phellinus (Poria) weirii has been identified in the subalpine forests of Japan (Aoshima 1953), the coastal Douglas-fir region of Oregon, Washington, and British Columbia (Childs 1970), and possibly in Douglas-fir plantations in France (Boullard 1972). Its history in the Pacific Northwest has been almost as varied as its distribution.

Phellinus weirii was first described from a section of decaying western red cedar in Idaho by W. A. Murrill (1914). Murrill gave the fungus two names: Fomitiporia weirii in honor of the collector J. R. Weir, and Poria weirii, following Saccardo's nomenclature. The fungus fit Saccardo's concept of Poria by its resupinate sporophore, globose spores, and growth on wood (Rea 1922). With such loose guidelines, Poria became a residual taxon to which practically any resupinate polypore was assigned (Domanski et al. 1973). The fungus remained in this genus for 60 years, until better data on meaningful characters were collected. In the early cultural descriptions by Overholts (1931) and Mounce et al. (1940), setal hyphae were observed to be embedded in walls and often projected into sporophore tubes. R. L. Gilbertson (1974) used this distinctive setal characteristic to reclassify the fungus as Phellinus weirii (Murr.) Gilbertson. This name is now accepted for the causal fungus of laminated root rot.

The hosts of P. weirii have long been debated. Murrill (1914) stated the host to be western red cedar (Thuja plicata) only, and Hubert (1927) agreed. While P. weirii was believed, until the mid-1940s, to attack only this species (Hubert 1931, Eades 1932, Boyce 1938, and Baxter 1943), evidence of its occurrence on other western

softwoods began to appear. On November 30, 1929 C. Riley found P. weirii decay on 18-year-old Douglas fir in British Columbia, but this observation was not published until 1940, when Mounce et al. (1940) reported P. weirii infection in Douglas-fir stands 18- to 200-years-old. This report caused forest pathologists to look more closely at decay in forests of the Northwest. The first indication of P. weirii in the Coast Range of Oregon was reported by Dr. T. S. Buchanan (1948), who had collected fungus from basal decay of 180-year old Douglas-fir in Coos County.

In 1948, Weyerhaeuser Timber Company reviewed its field survey for logging planning and noticed extensive openings in the stands, a few greater than four hectares, with chlorotic trees and dead standing snags (Sommer 1950). However, the Douglas-fir bark beetle (Dendroctonus psuedotsugae) was active in these dead and weakened trees, and the beetle was blamed for the stand's condition. The winter of 1949-1950 was blustery, and by spring extensive windthrow was noted in the Coos Bay region. In the summer of 1950, Dr. T. Childs, a U.S. Forest Service Pathologist, surveyed windthrown trees and noted P. weirii on practically every one. This discovery prompted Weyerhaeuser foresters to recheck beetle kill areas. They found almost every down tree was infected with P. weirii and most adjacent trees had incipient decay (Sommer 1950). Extensive infection was also reported in British Columbia (Wallis 1953). So, in the early 1950s, Douglas-fir was considered an excellent host for fungus. In one of the first descriptive articles about P. weirii on Douglas-fir, Buckland et al. (1954) believed the fungus to be a different variety of that described on western red

cedar, because they found some macroscopic differences between Thuja and Pseudotsuga isolates of the fungus. Working in northern Idaho, Hubert (1950) collected P. weirii from several species of conifers, and in the late 1950s, P. weirii no longer was considered host specific to western red cedar. The forest pest leaflets printed by the United States (Childs 1960) and Canada (Wallis and Reynolds 1967) described P. weirii on several additional conifers.

Phellinus weirii has been isolated from Douglas-fir, western red cedar, mountain hemlock (Tsuga mertensiana), western hemlock (T. heterophylla), grand fir (Abies grandis), Pacific silver fir (A. amabilis), subalpine fir (A. lasiocarpa), Englemann spruce (Picea englemannii), Sitka spruce (P. sitchensis), western larch (Larix occidentalis), lodgepole pine (Pinus contorta), western white pine (P. monticola), and ponderosa pine (P. ponderosa) (Hubert 1950, Buckland et al. 1954, and Hadfield and Johnson 1977). However, the most susceptible hosts are Douglas-fir, Pacific silver fir, subalpine fir, grand fir, Sitka spruce, Engleman spruce, western larch and western hemlock (Hadfield and Johnson 1977, Wallis 1976).

Phellinus weirii is considered a disease of the site because infection centers result from contact between living roots and infected roots and stumps left from the preceeding stand (Childs 1960). These infection centers range from a single tree to half a hectare or more in size and contain trees with various crown conditions-- from trees with full, green crowns to ones with yellowing and thinning crowns to standing snags or windthrown trees. In early stages of P. weirii decay the

roots and trunk show brown crescents in the cross section. As the disease progresses the wood breaks down into a loose mass which may completely disintegrate (Buckland et al. 1954). The rot is termed laminated because the annual rings tend to separate as the decay progresses. Tufts of brown setal hyphae usually occur in decayed wood and on root surfaces. Setal hyphae are diagnostic of P. weirii infection in stumps and roots.

Phellinus weirii is a lignin decomposer (Li et al. 1968b), and therefore a white rot fungus, though earlier accounts mistakenly classed it as a brown rot fungus (Hubert 1927, Eades 1932, and Baxter 1943). It utilizes glucose, xylose, and fructose (Li and Bollen 1975), which are common in the cambial region of Douglas-fir (Holmes and Kurth 1961). While it lacks the ability to reduce nitrate (Li et al. 1968a), it can use ammonium or amino nitrogen for protein synthesis (Li et al. 1967b).

Although P. weirii has survived in temperatures from -5° to $+30^{o}$ C. in the laboratory (Nelson and Fay 1974), optimal growth occurs from $+20^{o}$ to $+25^{o}$ C. (Li et al. 1967a, Nelson and Fay 1975). While Nelson (1971) reported $+7^{o}$ to 18° C. to be favorable for basidiospore germination, no work to date has related temperature to survival of the fungus in infected wood.

Phenolic compounds can inhibit growth of P. weirii. Barton (1967 1968 and 1972) found differences in extractives of healthy and P. weirii infected roots of Douglas-fir, and believed the compound extracted from healthy roots to be a phytoalexin. Morrison (1969) noted that hot water extracts of Thuja plicata heartwood inhibited growth of P. weirii from

other conifers. Li et al. (1969) proposed that the significant amounts of inhibitory compounds in red alder (Alnus rubra) tissue might account for its resistance to P. weirii. Further work has shown that red alder produces linoleic acid (Li et al. 1970) and phenolic acids (Li et al. 1972, Trappe et al. 1973) which inhibit growth of the fungus in vitro.

The method of spread by P. weirii has also been debated. Fomitopsis annosus, another root rotting organism, can spread by spores, which enter root systems through freshly cut stumps of thinned stands. Hubert (1927) believed that fire scars and injuries to the base of trees provided entry points for P. weirii. Buckland et al (1949) and Wright and Issaac (1956) found P. weirii associated with logging scars in western hemlock.

Thinning is important in management of Douglas-fir, so if spores did infect stumps, the coastal forests could be threatened. However, Nelson (1971) applied basidiospores of P. weirii to 21 freshly cut Douglas-fir stumps and observed no colonization by the fungus. The role of spores is not believed to be important. Instead, primary spread appears to occur when healthy root contact infected roots or buried sources of inoculum. Hubert (1950) found that in young stands of grand fir containing remnants of old western red cedars destroyed by fire, young fir established after the fire developed P. weirii decay. Childs (1963) associated newly infected Douglas-fir with residue from a previously infected stand.

With the root system providing a pathway, P. weirii can move through a stand with relative ease. Wallis and Reynolds (1962) observed P. weirii

growing 25 centimeters along the surface of roots in 12 months in sterilized culture jars. Nelson and Hartman (1975) measured an average annual rate for the opening of infection centers of 34 centimeters. Root grafts are not necessary for mycelial spread from tree to tree. Wallis (1957) traced recent infections of Douglas-fir to points of superficial contact of the root bark surfaces, and Childs (1963) cultured the same clone from a neighboring Douglas-fir and western red cedar. The root bark of Douglas-fir cannot stop hyphal penetration (Wallis and Reynolds 1965). P. weirii cannot grow through the soil alone. Wallis (1957) observed numerous roots passing within 20 millimeters of adjacent infected roots without showing any signs of infection, and of 2000 attempts to isolate P. weirii from 250 wood disks either sprayed with a soil suspension before inoculation or covered with soil during the incubation period, only one produced P. weirii (Nelson 1976a). It is believed that P. weirii competes poorly with soil microorganisms (Nelson 1969).

Though a poor competitor, it survives in stumps 20 years after cutting (Hansen 1976) and possibly 100 years (Childs 1963). This long survival may result from its ability to produce a boundary envelope or zone line of hyphae and inflated fungal cells (Garrett 1970). Nelson (1964 and 1967) found that P. weirii survived better in Douglas-fir heartwood cubes when it produced zone lines than when it had not. He believed this was a defense mechanism, because P. weirii did not form zone lines when soil microbial competion and conditions were favorable for its growth (Nelson 1973). Therefore, because of its ability to seal

itself from harm and survive in forest residue, P. weirii can remain a hidden danger for many years, waiting for the proper host and conditions to cause infection.

Phellinus weirii was not reported on Douglas-fir until the late 1940s, but by 1960 it had been reported on all Weyerhaeuser Timber Co. tree farms west of the Cascade Range in Oregon (Shea 1960), in 49 out of 361 National Forest plantations in Oregon and Washington (Johnson 1976), and in plantations in British Columbia (Johnson et al. 1972) by the mid-1970s. The annual loss by P. weirii in Oregon and Washington is estimated at about one million cubic meters (32 million cubic feet) (Childs and Shea 1967). The damage occurs as loss by windthrow, standing mortality, and declining growth of live infected tree.

Windthrow is a spectacular form of damage. Buckland et al. (1948) inventoried one stand of 80-year-old Douglas-fir in British Columbia and found 1700 windthrown trees, the majority caused by P. weirii decay.

Damage by the loss of expected grwoth of live infected trees is less obvious, but can be equally devastating. Buckland et al. (1954) measured a loss of 58 percent of the expected volume from 1935 to 1950 on a one-tenth-acre plot in a 30-year-old stand of Douglas-fir. Bier and Buckland (1974) observed a reduction of 21 and 47 percent of the expected increment of the dominant and co-dominant trees because P. weirii on two one-fourth acre plots. Hunt (1956) estimated about six percent of the stand in a 60-year-old Douglas-fir forest to be "destroy ed" by P. weirii, and Molner et al. (1961) found an average of 49 trees per hectare (20 per acre) to be diseased in a 50 year-old-stand. The

rate of damage is believed to double every 15 years (Childs 1970).

Measures to contain the fungus range from scarification (Weir and Johnson 1970) and trenching of infection centers (Wallis and Buckland 1955), to fertilization (Nelson 1976b) and species regulation (Trappe 1972) on these centers. Management proposals for infected stands by Childs (1970), Wallis (1976), and Hadfield and Johnson (1977) include stump removal, harvesting infected stands early, and the replanting of infected sites with resistant species. Russell (1971) recommends clear-cutting infection centers to avoid future losses from wind, ice and snow damage.

The Study Area

The Black Rock Forest Management Research Area (Black Rock Forest) is a part of the 405-hectare George T. Gerlinger Experimental Forest in Polk County, Oregon, four kilometers west of Falls City, and 64 kilometers northwest of Corvallis, Oregon. The 195-hectare (482-acre) research area is owned by the Oregon State Forestry Department, with research directed by Oregon State University's Department of Forest Science. Topography varies from nearly level to grades of 61 percent with elevations from 212 to 606 meters. The forest lies on the interior Coast Range and receives about 200 centimeters of precipitation annually, mostly as rain. The soils are lateritics developed from sandstone, shale, and silt stone of the Eocene Age, and igneous rocks of the Miocene Age. They have a granular structure and are moderately acid in the surface, becoming strongly acidic in subsoil horizons (Youngberg 1964).

The previous forest had an average of 104 Douglas-fir trees per hectare, and a basal area of 95.3 square meters per hectare (Berg 1977). This forest was logged between 1910 and 1918.

The present forest originated from natural seeding after logging and is 60- to 65-years-old. Most of the forest is Douglas-fir, with such associates as western hemlock (Tsuga heyerophylla), grand fir (Abies grandis), big leaf maple (Acer macrophyllum, bitter cherry (Prunus emarginata), western dogwood (Cornus nuttalii), and red alder (Alnus rubra). It can be classed as an Acer circinatum / Gaultheria shallon forest type (Franklin and Dryness 1973), with sites ranging from II to IV.

The Department of Forest Science used the forest for research on commercial thinning and the ecological changes that forest management causes in a forest. Most of the 64 plots were established and thinning treatments begun between 1953 and 1959. These 64 plots comprise 18.51 hectare and have thinning schedules varying from no thinning (control plots) to maintaining stand densities between 23 and 44 square meters. Thinning operations are done with horses or rubber-tired skidders.

Two areas were set aside on the forest to study P. weirii in 1961. One, the Upper Phellinus plot is 570 meters in elevation and consists of 27 one-tenth-acre subplots. The other, the Lower Phellinus plot is 303 meters in elevation and consists of 23 one-tenth-acre subplots.

Methods

In this study, trees were examined for infection by P. weirii, infected areas located and mapped, volume production measured, movement of the fungus recorded, and clonal studies undertaken on the Black Rock Forest.

Disease Diagnosis.

Since Wallis and Reynolds (1962) observed the development of surface mycelium in advance of infection of root wood, when distinctive surface mycelium was found on the root collar of a tree, it was designated as infected, and, if not, as not infected. Windthrown trees were examined for setal hyphae and lamination of roots, which Childs (1970) attributed to P. weirii. Phellinus weirii is such a poor competitor against other fungi that if it was present on a dead standing tree, P. weirii was considered the primary cause of death (Childs 1970).

Old-growth stumps are attacked by several decay fungi, so strict guidelines were based on Thomas and Thomas (1954), Boyce (1932), and Englerth (1942) to identify fungi present in them. These stumps were considered infected by P. weirii if setal hyphae were present or if the roots of the stump were hollowed out.

Survey and Mapping Procedures.

Known infected areas were located on a map of the Black Rock Forest by Alan Berg, Department of Forest Science. Subsequent field surveying located additional areas.

Stand maps were made of each infected area with a plane table and tape measure. Distances from the table to trees were measured and each

tree and old-growth stump was located on the map. Each tree, living or dead, was identified by number, and diameter at breast height, crown position, tree condition, and presence or absence of infection by P. weirii were recorded. Infected areas were mapped until a 10 meter band of uninfected trees completely encircling the infected ones was evident. The year of death was assumed to be 1978 if a windthrown tree retained its needles, and if standing dead trees had tight bark and retained fine branches. Otherwise, they were classed as older mortality.

The Upper and Lower Phellinus plots were mapped and each tree numbered and diameter recorded in 1961. When these areas were surveyed in 1978, copies of the original maps were used and subsequent disease incidence marked on them. Diameters, crown position, and conditions were also recorded.

In the office, these maps were redrawn and each tree designated either P. weirii mortality, alive and infected, or no surface mycelium found on roots, but within 10 meters of an infected tree.

Area and Volume Calculations

In the office, the area of infection centers and volumes of trees in the centers were calculated.

Infected areas were delineated by a line drawn on the maps half the distance between infected and uninfected trees to include all infected trees. A planimeter determined the area within this circle.

Areas out of conifer production contained brush fields, down trees, hardwoods, and conifers less than 20 centimeters in diameter too

suppressed to respond to release (Childs 1960, Berg personal communication). At age 60 a spacing of 7.3 by 7.3 meters provides for 185 trees per hectare, which is considered a favorable stocking level for Black Rock Forest (Berg personal communication). The line delineating areas out of production based on this spacing was drawn half the distance between living trees and dead ones and hardwoods, or 3.6 meters from living trees to encompass brush fields.

Cubic meter and board foot volumes for 1978 mortality and for living infected and uninfected trees were determined from local volume tables developed for Black Rock Forest (Berg 1967).

The Black Rock Forest control plots 7, 12, 16, 19, 21, 27, 37, 46 and 50 have received no silvicultural treatments and therefore provide a standard volume production in natural stands. To determine volume loss in cubic meters and board feet from P. weirii over time, each infected area was compared to an uninfected control plot that was close and similar to the infected area. By assuming that the volume on P. weirii - infected areas would have been the same as that produced on the control plots, an estimate was made of the volume that would have been on the infected area in 1978 had P. weirii not been present. The difference between this figure and the volume measured on the areas in 1978 is an estimate of the loss that has occurred over time on these infected areas.

Impacts on the Upper and Lower Phellinus Plots.

The Upper and Lower Phellinus plots were thinned in 1968 and 1972, and diameters of living, dead (due to P. weirii and other agents),

salvaged, and thinned trees recorded at those times. This survey provided diameters of living and dead trees in 1978. A local volume table (Berg 1967) provided volumes of these trees so that a volume loss over 17 years could be determined. The rate of loss over time was estimated by comparing the volume lost from 1962 to 1967 with the volume lost from 1968 to 1972 and from 1973 to 1978.

Using these records, total mortality and the percent of mortality caused by P. weirii was determined for 1967, 1972, and 1978. The percent of growth lost due to P. weirii was also calculated for these time periods. This percent was equal to the volume of trees killed by P. weirii divided by growth increase for that time period on each subplot.

The percent loss of gross periodic production from these areas from 1961 to 1978 was equal to:

$$\frac{\text{Total Volume of } P.\ weirii \text{ Mortality of All Subplots From 1961 to 1978}}{\text{Live Volume 1978} - \text{Live Volume 1961} + \text{Mortality Volume 1961 to 1978} + \text{Thinned Volume 1967, 1972}} \times 100 \text{ Of All Subplots}$$

Movement of P. weirii on the Upper and Lower Phellinus Plots.

The location of P. weirii mortality from pre-1961 to 1978 was marked on each subplot map, and circles drawn to include pre 1961, 1961, 1962 - 67, 1968 - 72 and 1973 - 78 kills.

The probability of a 1978 kill at distances from 0 to 6 meters (0 to 20 feet) from a 1961 P. weirii killed tree and a bound on the error were determined after Neter and Wasserman (1974) and Ott (1977).

A bound on the error contains these probabilities of a kill to within two standard deviations of their means. Three circles with radii equivalent to ground distances of 3, 4.5, and 6 meters (10, 15 and 20 feet) were centered on each 1961 kill and the number of trees killed by 1978 and the total number of trees within each radius counted. These proportions provide a point estimate of the probability of a tree being killed in 17 years and the interval range for 95 percent of the probabilities.

To determine an annual expansion of the pockets, distances from 1961 kills to trees killed in 1967, 1972, or 1978 were measured on the Lower Phellinus plot. This distance was divided by the number of years since 1961 to provide an average annual expansion rate.

Clonal Analysis.

Aspects of the history of infection centers can be reconstructed by clonal analysis. Phellinus weirii consists of many incompatible clones differing in growth rate, pH requirements, and other characteristics (Childs 1963). If two isolates of different clones are placed in the same petri dish, a dark line forms between colonies. If the isolates are from the same clone, the mycelium will merge with no dark line formation. Centers yielding different clones were once separate, but have merged over the years to become one center (Childs 1963).

Samples of wood from trees infected by P. weirii were collected from trees in infected areas A, H, K, and the Upper and Lower Phellinus Plots. Isolates from each tree were paired on malt agar plates, and the formation or not of dark lines recorded. Isolates were identified with the key by Nobles (1965).

Results

The Extent of *Phellinus weirii* on Black Rock Forest.

In this survey, 27 separate areas were mapped and inventoried (Figure 1). These areas varied from 0.02 hectare containing 5 trees to 0.71 hectare containing 148 trees. Altogether they comprise 7.13 hectares, which is 3.6 percent of the land area of Black Rock Forest. The results of the survey are summarized in Table I. All 27 infection centers contained living and dead Douglas-fir, some hemlock, grand fir and hardwoods. No evidence of *P. weirii* was found on grand fir, western hemlock or the hardwoods, though *Armillaria mellea* was also active on some of these areas. Of the Douglas-fir on these centers in 1978, 286 were killed by *P. weirii*, 1732 were infected with the fungus, and 2145 were not infected but were 10 meters or less from a source of *P. weirii*. Some old-growth Douglas-fir stumps were infected with *P. weirii*, but most were too deteriorated to evaluate.

Within 24 of the infection centers were areas considered nonproductive for confiers because they contained hardwoods, dead and down trees, brush fields, or trees to suppressed to significantly benefit from release. The combined area out of production due to *P. weirii* was 1.56 hectares.

There are *P. weirii* infection centers west of the Lower *Phellinus* plot. This area was not mapped or surveyed because of time limitations on the study. While there are no large nonproductive areas there, many trees are windthrown and as yet not salvaged. For the purposes of this study, that area is not considered a part of the Black Rock Forest, and its area is not calculated in the land area of the Forest.

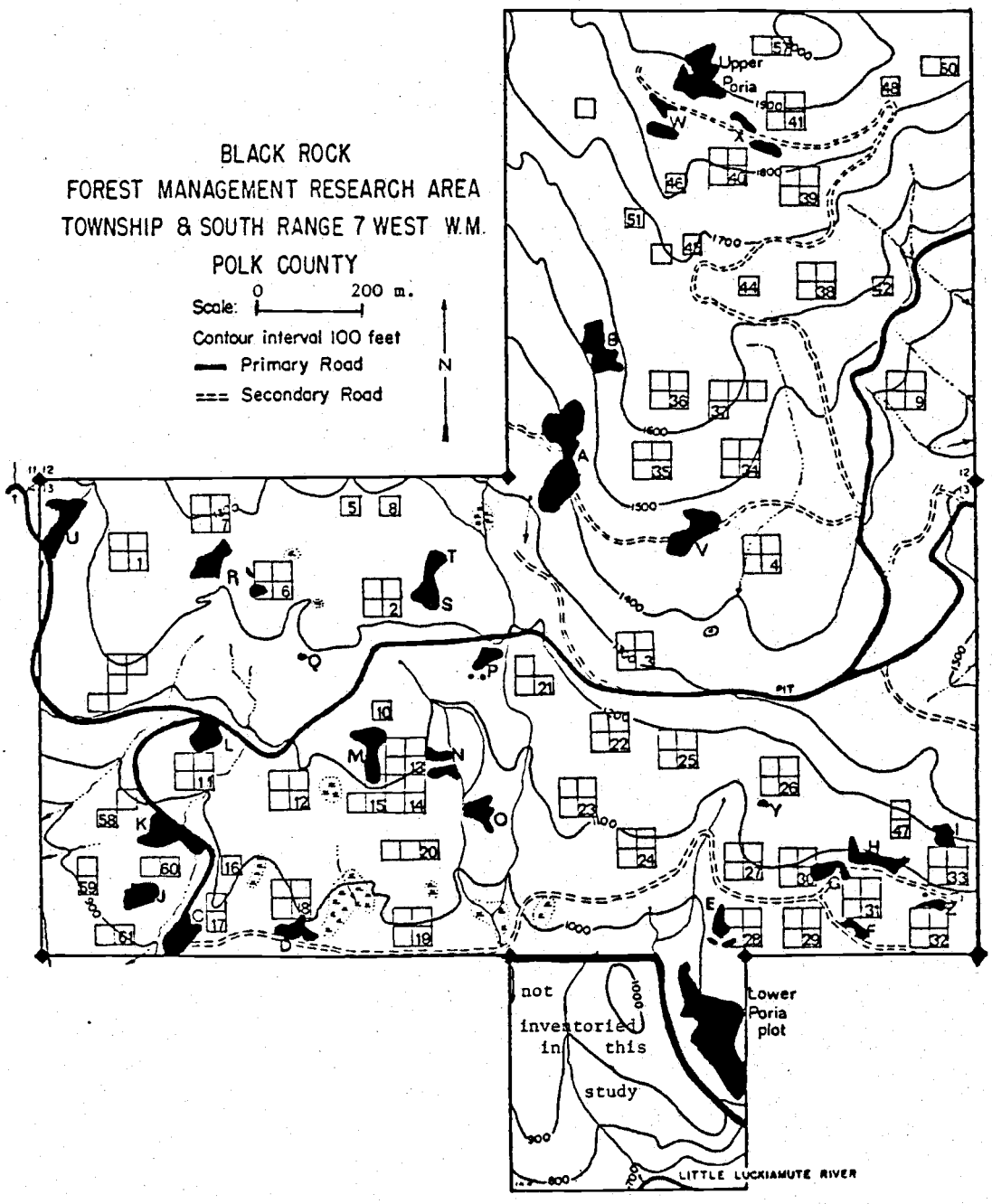


Figure 1. P. weirii infection centers on Black Rock Forest.

1978 Mortality Due to Phellinus weirii

The mortality of Douglas-fir caused by P. weirii is given in Table II. In one year 1978, P. weirii caused the death of 286 Douglas-fir with a total volume of 457.75 cubic meters (79,040 board feet). While most of the trees were windthrown, 30 percent of the killed trees were still standing at the time of inventory.

Estimated Cumulative Volume Loss Due to Phellinus weirii.

The nine control plots at Black Rock Forest (plots 7, 12, 16, 19, 21, 27, 37, 46, and 50) provide volumetric standards for the forest. As Table III shows, the 27 infection centers should have an estimated volume of 7095 cubic meters when control plots are used for a base estimate. However, the survey in 1978 only recorded 2981 cubic meters on these areas, which means over 65 years P. weirii has caused a net loss of 4114 cubic meters of Douglas-fir.

Cumulative losses in a managed stand would be greater. By the most productive thinning schedule, net production of volume has been estimated to increase an average of 10 percent over control plot volume (Berg 1978). Such an increase of the control plot base would mean that the 27 infection centers should have a volume of 7805 cubic meters as a result of thinning. With this as a new base volume, P. weirii could be estimated to have caused a cumulative volume loss of 4824 cubic meters over 65 years in a managed stand.

The removal of softwood volume and the growth of residual trees provided a measure of gross periodic production on the Upper and Lower Phellinus plots. P. weirii was responsible for a reduction by 7.3 percent of production on the upper plot and 14.4 percent on the Lower Plot in 17 years.

P. weirii was responsible for a loss in the volume growth on the Upper and Lower Phellinus plots. Tables IV and V record the percent of volume that was killed by P. weirii during the three time periods.

Variability of Mortality in the Upper and Lower Phellinus Plots.

Figures 2 and 3 illustrate the advance of mortality due to P. weirii on parts of the Upper and Lower Phellinus plots. The openings have not expanded evenly. Mortality has been erratic over 17 years. Both figures show living Douglas-fir scattered around the old killed trees, with few patterns of expansion present. The "doughnut effect" as explained by Hadfield and Johnson (1977) was not evident on these plots. In Figure 2, one group of trees show an expanding, though uneven, margin of mortality, while Figure 3 shows a more active expansion in another group. Group A in Figure 2 has trees killed to the south and southeast, while trees to the north and west have not died. Likewise, group B in Figure 3 has seen active expansion to the east while trees to the north have not been killed by the fungus. Both figures show groups of trees killed in 1961, but nothing has died since, even though living trees are close to the dead ones.

Records of the number of trees and their volumes killed since 1961 provide an example of the variability of loss due to P. weirii over 17 years. As summarized in Tables VI, VII, VIII, and IX, the number and volume of trees killed by P. weirii differs from year to year on the two plots. On the Upper plot, the number of trees killed between 1968 and 1978 was almost equal to the number killed between 1962 and 1967, but the volume lost by 1978 was 1.6 times the volume lost from 1962 to 1967 (Table VI).

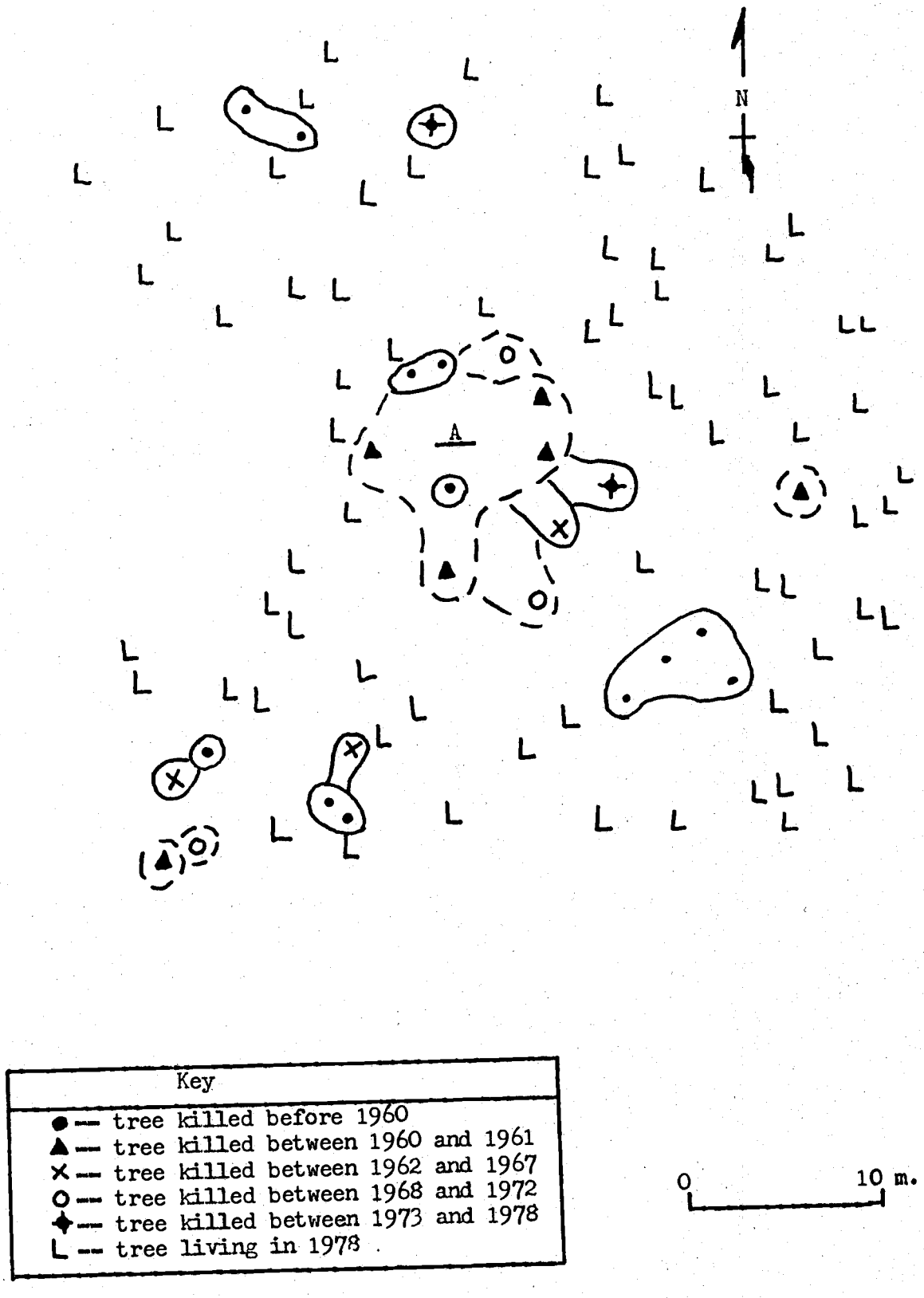


Figure 2. Approximate location of trees killed by P. weirii and those living from 1960 to 1978 on subplots 4-6 and 11-13 of the Upper Phellinus plot.

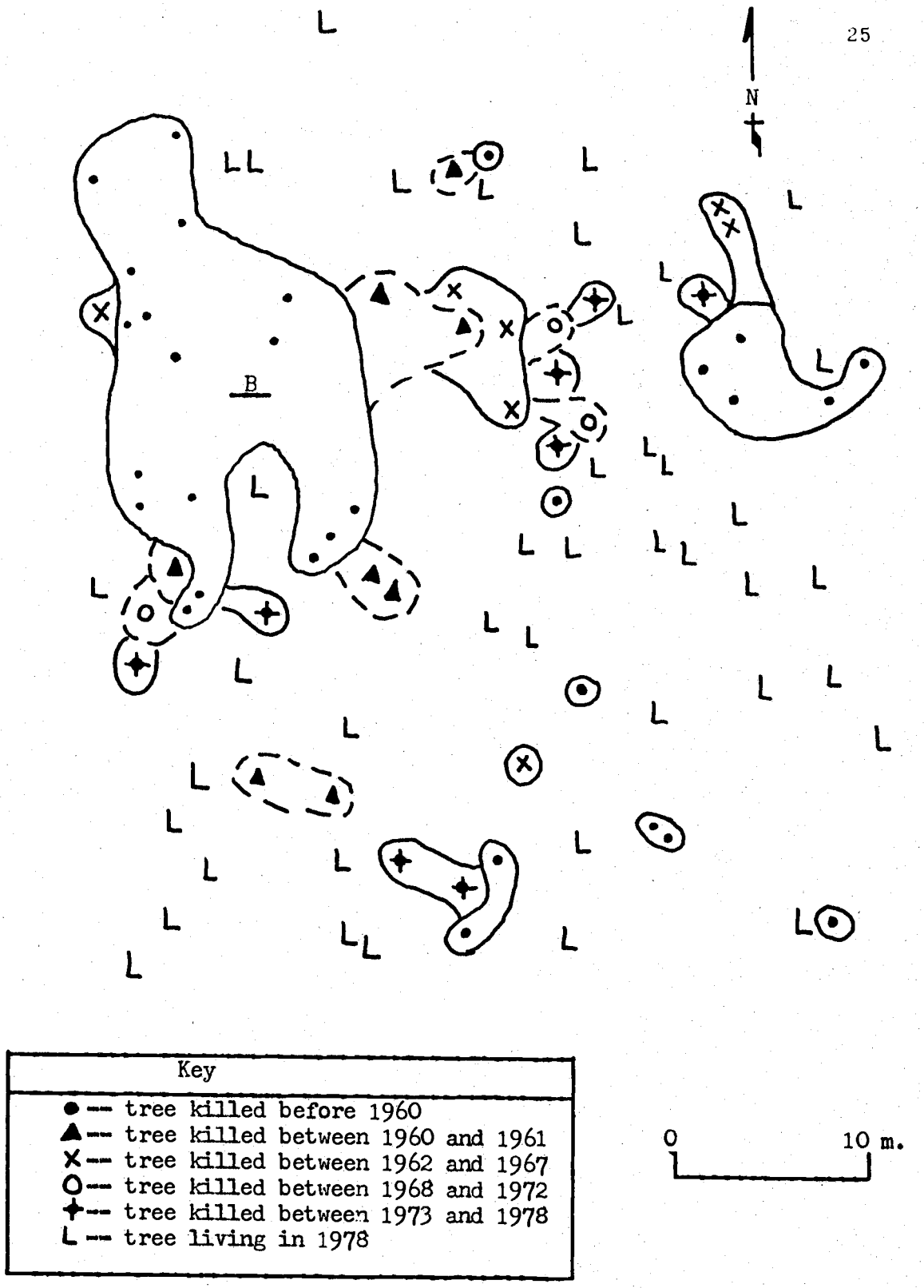


Figure 3. Approximate location of trees killed by P. weirii and those living from 1960 to 1978 on subplots 4-12 of the Lower Phellinus plot.

On the lower plot, the number of trees killed and the volume lost from 1968 to 1978 were 1.5 and 2.0 times that recorded from 1962 to 1967 (Table VII).

The loss from P. weirii is not consistent each year. Tables VIII and IX illustrate this by showing the amount and percent of total deaths that are attributed to P. weirii over 17 years. Of all the trees recorded as dead on the Upper Phellinus plot for the period of 1962 to 1967, 8 percent of them were killed by P. weirii while from 1968 to 1972, 11 percent and from 1973 to 1978, 10 percent were killed by P. weirii (Table VIII). In Table IX, P. weirii was responsible for 20, 18 and 56 percent of the dead trees by 1967, 1972 and 1978 respectively on the Lower plot.

Rate of Spread of Phellinus weirii Mortality at Black Rock Forest.

As shown in Table VII, the volume of trees killed from 1968 to 1978 on the Lower plot is 2.0 times the volume killed from 1962 to 1967. This was because although the number of trees killed did not double, individual trees grew larger and increased in volume. As shown in Table VIII, P. weirii was responsible for only 33 percent of the trees killed in 1978 on subplot 23, and yet had caused 89 percent of the volume loss that year. Many such examples are shown in Tables VIII and IX.

An even, linear movement of mortality was not ovserved. Table X summarizes a rate of mortality spread for seven trees in the Lower Phellinus plot. Earlier work by Nelson and Hartman (1975) and McCauley (1976) measured the average annual opening of the canopy to be 34 centimeters per year. These trees were measured because they were the only trees which expanded the opening of the infection center upon death. To

arrive at an average annual expansion of the opening the distance from a 1961 killed tree to the recently killed tree was divided by the number of years between 1961 and the year of death as recorded in the inventory. These measurements gave a variation of 4 to 77 centimeters per year for an expansion of the canopy opening, with an average of 29.6 centimeter per year for these seven trees.

The Upper and Lower Phellinus plots did not provide statistically reliable conclusions because of the small size of the samples. However, an analysis was made of the data to determine the point estimates and bound on the error which estimate the probability a tree 3, 4.5 or 6 meters from a 1961 kill had the probability of 0.11 ± 0.10, 1.17 ± 0.13, or 0.07 ± 0.07 respectively of being killed in 17 years.

Clonal Boundaries on Black Rock Forest

Isolate pairing on petri dishes showed antagonism between and within the five infection centers sampled. When one isolate from each of the five centers was paired with another, the clones showed antagonism to each other (Table XI). Because these five areas (A, H, K, Upper and Lower Phellinus plots) are scattered across the forest, it can be assumed that all infection centers on Black Rock Forest did not originate from a single P. weirii infection center.

Clonal differences were observed in pairings from within area A (Table XII). Three isolates from the center and southern portions of the area showed no antagonism toward each other, but when each of these was paired with an isolate from the northern portion of the area, all pairings produced dark lines. Such a reaction suggests area A is two separate areas that have merged (Childs 1963).

Four isolates from area H showed no antagonism toward each other, even though one was collected 30 meters north of what had previously been considered the edge of the infection center (Table XIII).

Table I. Inventory of Douglas-fir within 10 meters of infected trees on 27 infection centers (7.13 hectares) of *P. weirii* on Black Rock Forest in 1978.

Condition of Douglas-fir	No. of trees	Volume (cu. m.)	Volume (bd. ft.)
Killed by *P. weirii* in 1978	286	457.75	79,040
Living but infected by *P. weirii*	1732	2924.89	513,780
Living but not infected by *P. weirii* [a/]	2145	4692.38	853,555
Total	4163	8075.02	1,446,375

a/ these trees are within 10 meters of a source of *P. weirii*

Table II. Mortality of Douglas-fir in the year 1978 caused by *P. weirii* on Black Rock Forest.

Condition of the killed trees	No. of trees	Volume	
		cu. m.	bd. ft.
Windthrown	195	371.86	66,125
Standing dead	91	85.89	12,915
Total loss in 1978	286	457.75	79,040

Table III. Estimated volumes on *P. weirii* areas as natural stands over 65 years and as inventoried in 1978 on 7.13 hectares of Black Rock Forest.

Condition of the *P. weirii* area	Volume	
	cu. m.	bd. ft.
Potential volume as a natural uninfected stand [a/]	7095	1,214,700
Actual volume as inventoried in 1978	2981	523,400
Cumulative volume loss to date due to *P. weirii*	4114	691,300

[a/] based on control plot standards

Table IV. Percent of total production lost due to death by *P. weirii* on the Upper *Phellinus* plot.

Subplot No.	Percent production lost in: 1962-67	1968-72	1973-78
1	0	0	0
2	0	25.3	7.5
3	62.6	18.2	4.1
4	28.9	6.5	0
5	5.2	15.7	68.4
6	9.2	0	29.4
7	0	0	0
8	0	0	0
9	0	0	0
10	0	0	0
11	0	0	3.4
12	0	35.8	0
13	0	0	0
14	0	0	0
15	0	0	0
16	0	0	0
17	6.9	0	0
18	16.6	23.1	0
19	8.4	0	45.1
20	0	0	0
21	0	0	0
22	5.5	0	0
23	5.8	0	27.2
24	21.9	0	0
25	9.9	0	0
26	0	0	0
27	0	0	11.7
Total	6.4	4.3	7.2

Table V. Percent of total production lost due to death by *P. weirii* on the Lower *Phellinus* plot.

Subplot No.	Percent production lost in: 1962-67	1968-72	1973-78
1	10.4	0	0
2	0	0	0
3	0	0	93.8
4	0	0	165.0
5	0	0	17.3
6	0	0	0
7	51.8	25.5	29.9
8	59.6	53.7	136.3
9	58.2	0	28.9
10	0	0	0
11	53.8	62.3	0
12	0	0	0
13	0	5.2	101.5
14	0	0	0
15	54.6	0	64.5
16	0	0	0
17	0	0	64.6
18	14.9	0	0
19	0	0	0
20	0	0	0
21	0	0	0
22	0	0	0
23	0	0	0
Total	12.3	6.2	27.6

Table VI. Mortality due to *P. weirii* from 1962 to 1978 on the Upper *Phellinus* plot on the Black Rock Forest.

Time Period	No. of trees killed	Ave. DBH (cm.)	Total Volume (cu. m.)	Ave. Volume per tree (cu. m.)	Volume loss per year (cu. m.)
1962 - 1967	16	23.7	11.79	0.74	1.965
1968 - 1972	7	29.0	7.38	1.05	1.475
1973 - 1978	8	31.1	11.70	1.46	1.950
Total loss	31		30.87		

Table VII. Mortality due to *P. weirii* from 1962 to 1978 on the Lower *Phellinus* plot on the Black Rock Forest.

Time Period	No. of trees killed	Ave. DBH (cm.)	Total Volume (cu. m.)	Ave. Volume per tree (cu. m.)	Volume loss per year (cu. m.)
1962 - 1967	14	31.3	16.21	1.16	2.70
1968 - 1972	6	33.8	9.03	1.50	1.81
1973 - 1978	15	33.2	24.33	1.62	4.05
Total loss	35		49.57		

Table VIII. Amount and percent of mortality attributed to *P. weirii* on the Upper *Phellinus* plot in 1967, 1972, and 1978 inventories.

Subplot	Mortality due to *P. weirii* in:						Mortality volume (cu. m.) due to *P. weirii* in:					
	1962-67		68 - 72		73-78		1962-67		68-72		73-78	
No.	No.	%[a]	No.	%	No.	%	Vol.	%[b]	Vol.	%	Vol.	%
1	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	2	50	1	33	0	0	1.71	66	0.20	31
3	2	28	1	25	1	33	3.35	90	0.89	54	0.20	37
4	2	50	1	28	0	0	2.32	70	0.34	36	0	0
5	1	8	1	20	1	33	0.42	23	1.06	68	3.80	92
6	1	11	0	0	1	50	0.42	33	0	0	1.50	93
7	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	1	50	0	0	0	0	0.30	59
12	0	0	1	17	0	0	0	0	1.92	75	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0
17	1	57	0	0	0	0	0.42	63	0	0	0	0
18	2	20	1	33	0	0	1.35	50	1.46	84	0	0
19	1	8	0	0	1	14	0.67	47	0	0	2.60	67
20	0	0	0	0	0	0	0	0	0	0	0	0
21	0	0	0	0	0	0	0	0	0	0	0	0
22	1	20	0	0	0	0	0.31	43	0	0	0	0
23	1	8	0	0	1	33	0.42	22	0	0	2.30	89
24	3	18	0	0	0	0	1.44	45	0	0	0	0
25	1	33	0	0	0	0	0.67	74	0	0	0	0
26	0	0	0	0	0	0	0	0	0	0	0	0
27	0	0	0	0	1	20	0	0	0	0	0.80	51
Total	16	8	7	11	8	10	111.79	31	7.38	23	11.70	44

[a] percent of the total number of trees killed

[b] percent of the total mortality volume

Table IX. Amount and percent of mortality attributed to *P. weirii* on the Lower *Phellinus* plot in 1967, 1972, and 1978 inventories.

Sub-plot	Mortality due to *P. weirii* in:						Mortality volume (cu. m.) due to *P. weirii* in:					
	1962-67		68-72		73-78		1962-67		68-72		73-78	
No.	No.	%[a]	No.	%	No.	%	Vol.	%[b]	Vol.	%	Vol.	%
1	2	25	0	0	0	0	0.63	17	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	1	50	0	0	0	0	3.50	93
4	0	0	0	0	2	100	0	0	0	0	6.31	100
5	1	25	0	0	1	50	0.68	49	0	0	0.61	77
6	0	0	0	0	0	0	0	0	0	0	0	0
7	1	100	1	33	1	100	2.58	100	0.58	90	0.61	100
8	4	67	2	66	4	100	3.54	90	3.19	98	4.82	100
9	1	50	0	0	2	100	2.31	96	0	0	0.97	100
10	0	0	0	0	0	0	0	0	0	0	0	0
11	2	50	1	100	0	0	1.90	85	2.74	100	0	0
12	0	0	1	100	2	100	0	0	2.18	100	3.22	100
13	0	0	1	50	0	0	0	0	0.34	67	0	0
14	0	0	0	0	1	50	0	0	0	0	2.88	87
15	2	14	0	0	0	0	3.39	54	0	0	0	0
16	0	0	0	0	1	100	0	0	0	0	2.01	100
17	0	0	0	0	0	0	0	0	0	0	0	0
18	1	12	0	0	0	0	1.18	43	0	0	0	0
19	0	0	0	0	0	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0	0	0	0	0	0
21	0	0	0	0	0	0	0	0	0	0	0	0
22	0	0	0	0	0	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	0
Total	14	20	6	18	15	56	16.21	42	9.03	41	24.33	90

[a] percent of the total number of trees killed

[b] percent of the total mortality volume

Table X. Rate of spread for *P. weirii* mortality based on time and distance from 1961 kills on the Lower *Phellinus* plot.

Subplot No.	Tree No.	Distance from a 1961 killed tree or stump (cm.)	Time between 1961 and the year of death of the tree (years)	Rate of Spread (cm./yr.)
1	17	154	7	22
3	8	461	17	27
4	12	517	17	30
5	34	538	7	77
7	2	461	11	42
7	4	77	17	4
7	30	38	7	5
			Average	29.6

Table XI. Incidence of line formation (+) and no line formation (--) between paired isolates of P. weirii from five separate infection centers on Black Rock Forest.

		Source of Isolate				
		A	H	K	Upper Phellinus plot	Lower Phellinus plot
Source of Isolate	A	--	+	+	+	+
	H	+	--	+	+	+
	K	+	+	--	+	+
	Upper Phellinus plot	+	+	+	--	+
	Lower Phellinus plot	+	+	+	+	--

Table XII. Incidence of line formation (+) and no line formation (--) between isolates cultured from four trees infected by *P. weirii* on infection center A.

		Tree Number			
		133	214	250	390
Tree Number	133	--	--	--	+
	214	--	--	--	+
	250	--	--	--	+
	390	+	+	+	+

Table XIII. Incidence of line formation (+) and no line formation (--) between isolates cultured from two trees and two old stumps infected by P. weirii on infection center H.

		Stump No.		Tree No.	
		115	177	95	117
Stump No.	115	--	--	--	--
	177	--	--	--	--
Tree No.	95	--	--	--	--
	117	--	--	--	--

Discussion

The Impact of *Phellinus weirii* on Black Rock Forest.

The damage *Phellinus weirii* caused on Black Rock Forest was in the form of mortality of growing stock, with a reduction in volume production and the formation of nonproductive forest land.

Only 3.6 percent of the land area contained trees infected by *P. weirii*, but 286 Douglas-fir with a combined volume of 475.75 cubic meters or 79,040 board feet were lost in 1978 alone. Such a loss in one year, although it does not occur every year can reduce the stocking severely in some areas of the forest. With a final thinning in 1979, continued loss of this magnitude in the future disrupts planned yield because trees killed would be expected to be present at the time of the final harvest.

Although both grand fir and western hemlock are susceptible to *P. weirii* (Hadfield and Johnson 1977), they were free of infection in this survey. Only seven grand fir were within 10 meters of a source of *P. weirii* and had an average diameter of 17 centimeters, which was less than the average of the Douglas-fir on the sites. The hemlock surveyed were all reproduction with diameters no larger than 10 centimeters. While the hemlock were scattered throughout the infection centers, their roots probably have not come into contact with *P. weirii* inoculum yet.

The loss that occurs over the life of the forest is cumulative. This fungus does not act like a bark bettle or fire, which causes great loss in one or a few seasons. It is a part of the ecosystem and causes death at anytime after it becomes established on roots of Douglas-fir.

As shown in Table III, P. weirii has caused a reduction of growing stock over 65 years. The 7.13 hectares of infected area should have 7095 cubic meters of Douglas-fir (based on standards from unthinned control plots), but the survey in 1978 recorded only 2981 cubic meters: a cumulative loss of 4114 cubic meters has occurred over the life of the stand. This cumulative loss does not necessarily mean total loss. Salvage operations have removed many of these killed trees, but such losses affect stand productivity.

Phellinus weirii was estimated to reduce gross periodic production of the Upper Phellinus plot by 7 percent and that of the Lower plot by 14 percent in 17 years. The Lower plot is about five years older, and on a mesic site. Its Douglas-fir are larger, so a reduction in volume is larger when they die than on the Upper plot, which is a xeric site with smaller, more crowded stems. Berg (1978) estimated a 10 percent increase in net production of managed stands over unmanaged stands in 23 years under the best thinning regime at Black Rock Forest. If P. weirii can cause a net reduction between 7 and 14 percent on selected areas, gains from thinning may soon be lost.

P. weirii also has had an impace on the volume of these two plots. As shown in Tables IV and V, the fungus has killed a high percentage of the volume on some subplots. As shown in Table IV, subplot 3 lost 62.6 percent of its gross periodic volume to P. weirii from 1962 to 1967, and subplot 5 had 68.4 percent of its volume increment from 1972 to 1978 killed by the fungus. On subplots 7 and 8 of the Lower plot, P. weirii had destroyed much of the stand's volume (Table V). These figures are similar to growth loss reported by Bier and Buckland (1974),

who measured 21, 47, and 158 percent of the expected growth to be killed by P. weirii on 3 one-fourth-acre plots in six years. Buckland et al. (1954) reported 27 percent of the expected volume lost from 1935 to 1940, 13.1 percent from 1940 to 1945, and 34.1 percent from 1945 to 1950 on a one-tenth-acre plot of Douglas-fir in British Columbia. Although this total reduction in growth may seem small when included in the stands as a whole, the destruction within a single subplot can be total. Subplot 7 of the Lower Phellinus plot has had much of its volume reduced by P. weirii, and only had three Douglas-fir standing on the plot in 1978.

The impact of P. weirii has varied from year to year. As shown in Tables VI and VII, the Upper and Lower Phellinus plots have had periods of time with much mortality followed by years of few deaths. From 1968 to 1972, the Upper plot lost only seven trees to P. weirii after losing 16 trees from 1962 to 1967. The Lower plot varied in a reverse way, with more stems lost from 1973 to 1978 than from 1968 to 1972. Records from salvage operations, summarized in Table XIV show such variation from year to year over the whole forest. This is a table of trees that were salvaged each year, but an attempt was made to salvage all killed trees over 17 centimeters in diameter and does reflect the number of trees killed by P. weirii each year. Years such as 1958, 1972, and 1974 had minimal damage, while large number of trees were killed in 1962, 1968, and 1978. With such variation in damage, management proposals for infected stands must be flexible.

Only 1.56 hectares are totally nonproductive for conifers due to P. weirii, which is 0.8 percent of the land area of Block Rock Forest.

Table XIV. Number and volume of Douglas-fir killed by P. weirii and salvaged from 1957 to 1978 on the Black Rock Forest (Berg 1979).

Year	No. of trees	Volume			
		Total (cu. m.)	per tree (cu. m.)	Total (bd. ft.)	per tree (bd. ft.)
1957	24	34.61	1.44	7,020	292.5
1958	15	17.49	1.17	3,380	225.3
1959	84	102.70	1.22	15,890	189.2
1960	24	25.27	1.05	4,488	187.0
1961	118	108.23	0.92	17,795	150.8
1962	180	195.99	1.09	34,600	192.2
1963	30	27.15	0.90	4,390	146.3
1964	52	81.29	1.56	15,385	295.9
1965	85	83.97	0.99	13,700	161.1
1966	26	33.89	1.30	5,985	230.2
1967	no data available				
1968	171	224.16	1.31	38,985	227.9
1969	68	116.47	1.71	21,390	314.6
1970	27	41.97	1.78	7,510	278.1
1971	92	136.82	1.49	24,975	271.5
1972	20	42.53	2.13	7,870	393.5
1973	31	45.72	1.47	7,845	253.1
1974	13	28.35	2.18	5,170	379.7
1975	46	104.66	2.27	18,500	402.2
1976	57	145.52	2.55	28,002	491.3
1977	no data available				
1978	140	282.64	2.02	50,655	361.8
Total	1303	1879.43		333,535	

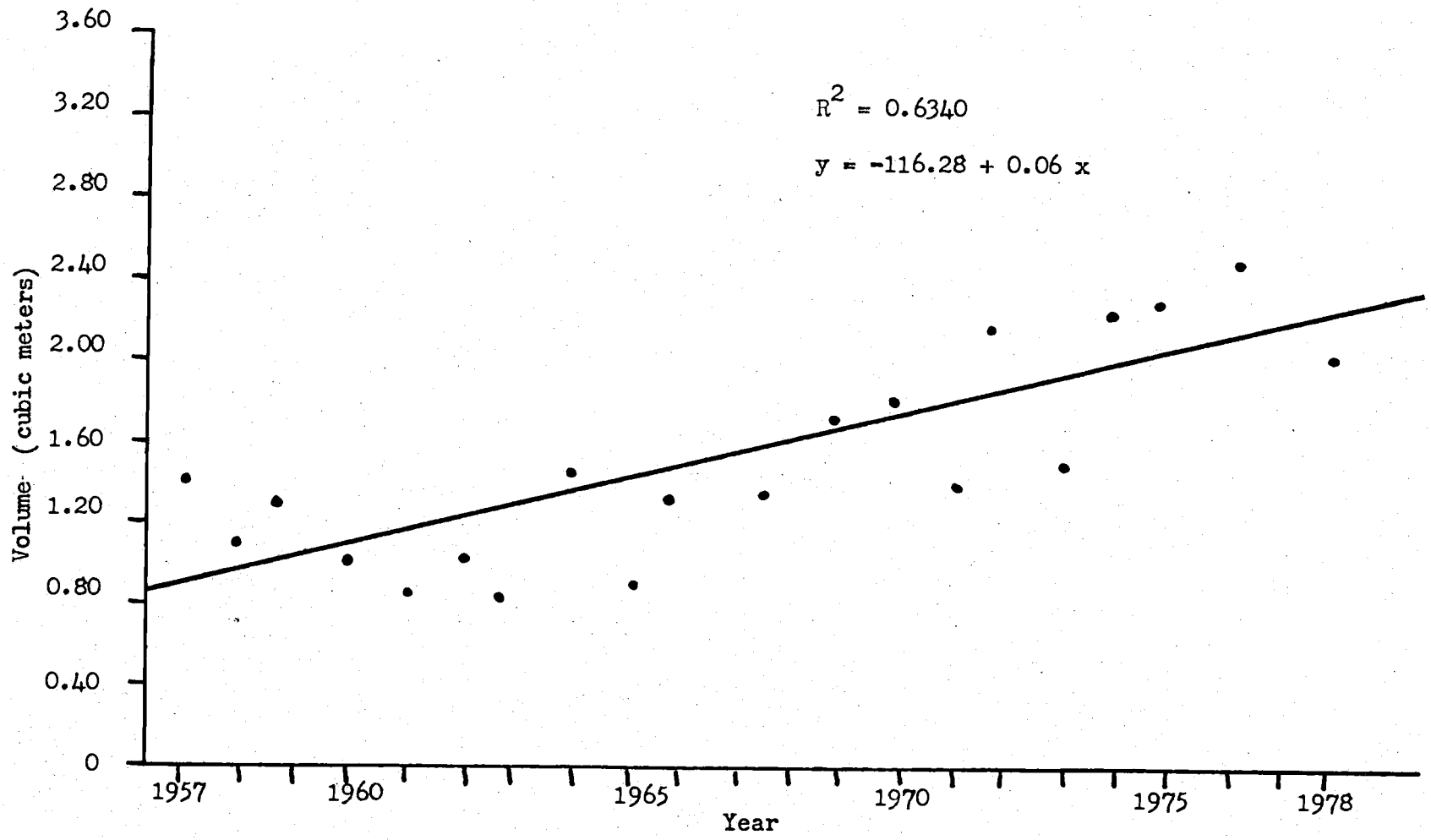


Figure 4. Relationship between year and cubic meter volume per tree of salvaged Douglas-fir killed by P. weirii on the Black Rock Forest between 1957 and 1978.

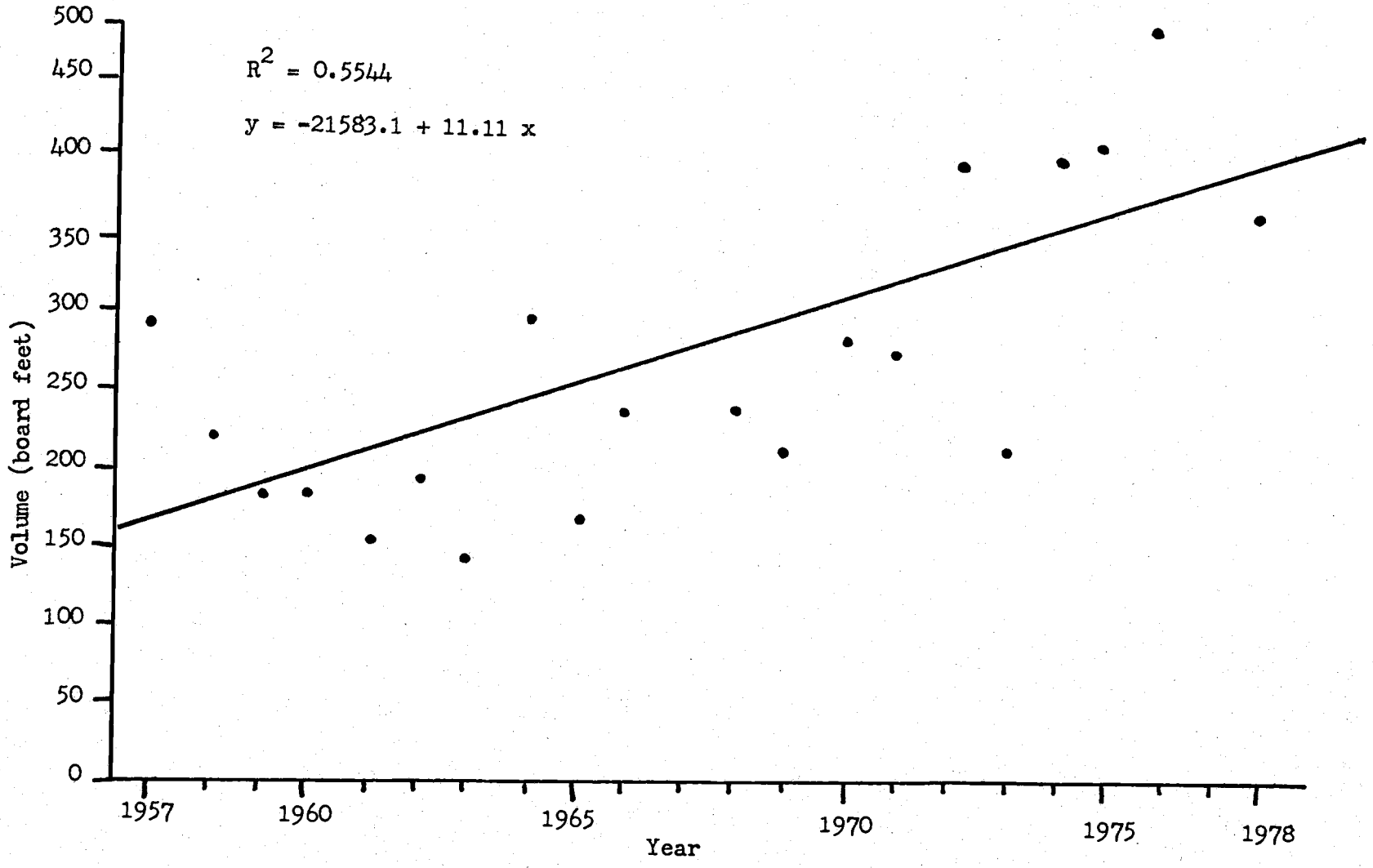


Figure 5. Relationship between year and board foot volume per tree of salvaged Douglas-fir killed by P. weirii on the Black Rock Forest between 1957 and 1978.

These nonproductive areas have thus not had a great impact on the the forest. The openings are occupied by bracken fern (Pteridium aquilinum), Oregon grape (Berberis nervosa), and salal (Gaultheria shallon), which retard reproduction of Douglas-fir.

Dynamics of Phellinus weirii Movement Through a Forest Stand.

The occurrence of mortality is dependent on two factors: occurrance of wind storms and movement of the fungus along root surfaces. Wind storms occurring during 1978 perhaps were a primary reason that 195 Douglas-fir were windthrown. Some years do not have severe winds. In 1964 only 4 trees with P. weirii were windthrown, and in 1974 8 Douglas-fir were windthrown (Berg 1979).

Although wind storms blow down infected trees, Douglas-fir are killed and infection centers enlarged primarily by fungal movement from tree to tree. No attempt was made to delineate the yearly movement of mycelium from tree to tree, but records of trees killed from 1962 to 1978 on the Upper and Lower Phellinus plots illustrate expansion of mortality. As seen in Figures 2 and 3, mortality has not expanded uniformily. Some openings have enlarged in an irregular pattern, while others have not expanded at all in 17 years.

Mortality centers have not expanded at an even rate on Black Rock Forest during the period investigated. One center on the Lower plot expanded at rates ranging from 4 to 77 centimeters per year (Table X). Figures 2 and 3 show that death passed by some trees only to kill others further out and cause an uneven margin of mortality. Clonal analysis show some infection centers composed of different clones,

which Childs (1963) attributed to the merger of separate infection centers into one center over time. Infection center F had only one tree killed by the fungus in 1978 even though three Douglas-fir were no more than 4 meters from several old infected windthrown trees. Infection center K had two windthrown trees in 1978 in an area with no evidence of old mortality and 45 meters from recently killed trees.

Even though expansion of infection centers does not occur at a constant rate, spread of the fungus can be categorized by the increase in the level of damage over time. Childs (1955) first theorized that the rate of damage due to P. weirii would double every 10 to 20 years. This theory has evolved to damage doubling every 15 years (Childs 1970) and the number of trees killed doubling every 15 years (Hadfield and Johnson 1977). If we assume that by "damage" Childs (1970) meant the volume lost, then this type of damage has doubled on the Lower plot. As shown in Table VII, the volume lost from 1962 to 1967 was 16.21 cubic meters. Double this value and we now have an estimated loss of 32.42 cubic meters by 1982. The combined volume loss from 1968 to 1978 on the plot was 33.36 cubic meters. A doubling of the volume lost has occurred in 11 years, however, such a loss did not occur on the Upper plot (Table VI).

Childs (1955) first proposed that the rate of damage would be the value that had doubled. The volume loss per year has remained almost constant on the Upper plot (Table VI), and increased only slightly on the Lower plot (Table VII). Both the plots had almost the same number of trees killed during each time period, which does not follow the theory that the number of trees killed would double every 15 years (Hadfield and Johnson 1977).

Both the average diameters of the killed trees and the average volume per tree increased from 1962 to 1978 on the Upper and Lower Phellinus plots. Such a trend for the whole forest is shown in Figures 4 and 5. But even with this increase in tree size, the rate of damage on the two plots, as measured by the volume loss per year, has not greatly increased over time. Even though the Upper plot had almost exactly the same volume loss per year in each time period, it only took half the trees from 1973 to 1978 to maintain the rate obtained from 1962 to 1967. On the Lower plot, the reverse was true. The number of trees killed from 1973 to 1978 was almost the same as that killed from 1962 to 1967, and yet the volume loss per year was higher for the former period than for the latter one. The rate of damage has increased over 11 years on both plots, but at a slower rate than theorized by Childs (1970). The loss on the Upper plot from 1968 to 1978 was only 0.88 times the volume loss per year from 1962 to 1967. On the Lower plot, the volume loss per year from 1968 to 1978 was 1.12 times the loss from 1962 to 1967.

The Impact of Phellinus weirii on Managed and Unmanaged Forests.

The impact of P. weirii differs with the level of management of infected stands. An unmanaged stand tends to lose more over time because most killed trees are not salvaged. In managed stands, salvage operations allow recovery of much of that loss. Even though in Table III P. weirii is estimated to have caused the death of 4114 cubic meters of Douglas-fir, salvage records show that 1879 cubic meters or 46 percent of that estimated loss have been salvaged since 1957 (Table XIV).

The stands have been entered often enough to minimize loss through decay of the mortality.

However management practices implemented to increase growth of stands may increase losses on P. weirii infected stands. Thinning operations may increase windthrow because infected trees cannot withstand the loss of support from neighboring trees (Wallis 1954). Thinning operations on the Black Rock Forest increased net production by 10 percent over unthinned control plots in one test (Berg 1978). Infection centers with such an increase would have an estimated volume of 7808 cubic meters of Douglas-fir. With only 2981 cubic meters inventoried on these centers in 1978, the cumulative loss has increased from 4114 to 4827 cubic meters. Salvage operations may recover volume loss of growing stock over the rotation, but they cannot alleviate the cumulative loss of growing stock documented by the Upper and Lower Phellinus plots.

Management Recommendations for Infected Stands.

Management proposals for forests infected with P. weirii were ineffective at first because the biology of the fungus was poorly understood. Sommer (1950) relates that Weyerhaeuser Timber Co., becoming aware of P. weirii in some of its stands of Douglas-fir, began salvage operations to remove windthrown trees. Childs and Clark (1953) reported windthrown timber usable for saw timber even after lying on the ground five years, so the operations were able to pay for themselves. Unfortunately Weyerhaeuser Timber Co. found a lot of brush in the openings of the infection centers, so the Company planned to fell infected

trees and skid them through these brush fields to break up the brush and make the site favorable for replanting with Douglas-fir. If this program were carried out, Douglas-fir seedlings would be planted in the midst of sources of P. weirii ranging from old-growth stumps to roots of recently infected trees. This program provided the fungus with a new food base and prepared the new stand for future deaths.

As the biology of the fungus became better understood, especially its ability to survive in roots for up to 100 years (Childs 1963), management proposals dealt mostly with control of the disease. The proposals included clearcutting of infection centers plus removal of a one-tree- wide ring around the center (Buckland and Wallis 1956), and the replacement of susceptible with resistant species on these sites to allow the fungus to die out for want of food (Child 1955). Current proposals have not changed much.

The first step toward management of infected stands is to locate the infections. Weir and Hubert (1918) emphasized the need to map disease distribution to determine pathological rotations as a basis for harvest schedules. Today, in the Cascade Mountains of Oregon (Johnson and Wear 1975) and in northern Idaho (Williams and Leaphart 1978), aerial photography has proven effective for identifing and mapping P. weirii infection centers. This tool has yet to be perfected for use in the Coast Range. Aerial photographs of Black Rock Forest do not clearly show the infection centers mapped in 1978.

Management options for infected stands from Childs (1970), Wallis (1976), and Hadfield and Johnson (1977) are:

1) become familiar with the fungus and be able to identify it in the forest,

2) remove stumps and deep rake the ground to break up roots in infection centers,
3) do not thin within 50 feet (15 meters) of infection center edges,
4) harvest infection centers early,
5) clearcut infection centers and them implement one or more of the following options:
   a) keep all susceptible species off the site and allow brush to develop in the centers,
   b) do not plant a susceptible species within 100 feet (30 meters) of an infection center, or
   c) replant those centers with resistant species.

The first option is needed to carry out any or all of the other options. In this survey, trees with surface mycelium were often found up to 24 meters (79 feet) from the nearest killed tree. If infection centers are delineated only by visible signs such as windthrows and crown characteristics, much of an infection center will be missed by any treatment.

Not all of these options would likely be effective on Black Rock Forest. Removal of infected stumps and deep raking of roots will remove much of the inoculum and reduce the hazard of infection to the future stand (Weir and Johnson 1970). But such an operation would require removal of at least 286 stumps and root portions at Black Rock Forest in 1978 alone. Perhaps the author thought this operation could be performed during the skidding of the timber, which would not require a special trip by the equipment and reduce the cost. But we need to take into consideration the ability P. weirii has to sustain itself in buried roots for long periods of time. If we are not thorough in the cleaning and raking operation, inoculum will still be on the site.

Old-growth stumps may be too decayed to be uprooted, but they serve as a reservior of P. weirii (Childs 1963). Also, this option depends on the topography. Root raking and soil disturbance on steep slopes could cause erosion problems.

Wallis (1954) found heavy losses due to windthrow after thinning, so the recommendation to not thin within 50 feet (15 meters) of a center's edge may reduce losses during the rotation. The current policy at Black Rock Forest is to leave infection centers alone except to salvage mortality. Such a policy has allowed most infection centers to contain trees with diameters of 70 centimenters or more.

The option to harvest infected stands early requires careful evaluation before implementation. This practice is recommended because it would allow salvage of recently killed trees and living infected trees before serious decay loss. But such a policy depends on condition and age of the stand. The Douglas-fir on Black Rock Forest provide enough volume to pay for the salvage operation, but if the stand is young, the operation might be uneconomical. To pay costs, several healthy trees beyond an infection center might have to be logged, which in the long run deprives that stand of potential volume from those trees.

If brush is allowed to develop in clearcut infection centers for about 40 years, P. weirii might die out of the site because of lack of food. However, brush will not lessen viability of P. weirii in old-growth and second-growth stumps, let alone mycelium on living trees surrounding the brush fields. To allow land, which can support other tree species, to remain fallow for 40 years is not economically sound.

If the infected stand contains trees of merchantible size, then the area could be clearcut and treatment of the infected site considered. Because the fungus grows along root surfaces, harvesting a ring of healthy Douglas-fir around an infection center might contain the fungus within that area. Such an operation was carried out on infection center H in 1958. The center as it appeared at that time was clearcut with a band of apparently healthy trees 1 to 2 trees wide around the center. In 1978, two windthrown trees 8 to 32 meters beyond this protective band yielded isolates which displayed no antagonism to isolates from two stumps cut in 1958 (Table XIII). Either the band of stumps was not an effective barrier to fungal movement or the extent of the infection center was not completely known at the time of the treatment. The former reason points to the inadequacy of the treatment on this center, while the latter stresses the importance of identifing the boundaries of infection centers before treatment.

If we do effectively delineate boundaries of infection centers and then clearcut, the areas must not be replanted with susceptible species such as Douglas-fir, mountain hemlock, grand fir, or Pacific silver fir. Planting susceptible tree species in infected sites is almost a guarantee that many of those trees will not reach biological or economic maturity, and that the volume lost over the rotation can be large. Wallis (1976) and Hadfield and Johnson (1977) proposed that no susceptible species be planted within 100 feet (30 meters) of infection centers. This practice will take more land out of production of desirable species. However, such a policy on Black Rock Forest would consolidate several individual centers into more manageable sizes: the number

of separate centers would drop from 27 to 15, and the area involved would increase from 7.13 to 15.3 hectares.

Planting infection centers with species intermediately susceptible to P. weirii may not result in serious loss, but they might support the fungus on the site longer than if it was planted with a resistant species.

Management Recommendations for Black Rock Forest.

Phellinus weirii will continue to have an impact on the management and yield of stands on the Black Rock Forest. Extending the regression line equations in Figures 4 and 5 the average volume loss in 1998 will be 3.4 cubic meters (617.5 board feet) per tree, or 1.56 times the loss in 1978. While salvage operations will not allow this to be a total loss, such an increase over time will demand action by the forest manager.

The options open to the forester are to ignore the disease, liquidate the stands immediately and replant with less susceptible species, or carry out salvage operations throughout the rotation. The trends of damage on the Upper and Lower Phellinus plots show that P. weirii cannot be ignored. Mortality has occurred throughout the life of the forest, with 1978 a particularly damaging year for Douglas-fir.

Because Black Rock Forest is 60- to 65-years-old, its timber is valuable enough to justify harvesting of infected stands now. If these centers were cleared along with an outer band of trees, the areas would have to be planted with trees resistant to the fungus. Resistant species ecologically suited to the forest include western red cedar, incense cedar, big leaf maple, red alder, Pacific madrone and cottonwood (Hadfield and Johnson 1977).

If full harvest of these stands is not desirable, salvage operations should be carried out until the end of the rotation. Black Rock Forest has such a network of roads that killed trees can be salvaged easily and economically. Salvage operations can be carried out by the owner or by a contract logger. Such a program would minimize losses and provide income to the owner.

For Black Rock Forest, the continuation of salvage operations with yearly mapping of P. weirii mortality is recommended. Because it is an experimental forest, there is no set rotation age, but when areas are finally harvested, a clearcutting of infection centers and the subsequent replanting of commercially valuable resistant species would be necessary to limit future losses from this fungus.

Conclusions

Phellinus weirii has had an impact on the productivity and growing stock of Black Rock Forest over the life of the forest. Since 1957, when records of its activity began, the fungus has been responsible for the death of over 1300 trees. Several infection centers have merged over this time. While only a small percentage of the forest contains infected trees, loss can be high on those areas, and the average volume of killed trees is predicted to increase each year.

Phellinus weirii plays a part in the regulation of the age, density, and species composition in Douglas-fir forests. This part has gained importance since Douglas-fir has achieved high status in the economy of the Pacific Northwest, but by trying to grow Douglas-fir continuously on land containing infected roots, the disease cycle is continued on these lands. As Pogo said so well, "We have met the enemy, and he is us." The replacement of Douglas-fir by other species on these infected areas is the only way presently to reduce Douglas-fir losses, and provide infection-free sites for succeeding generations of forests in the Pacific Northwest.

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